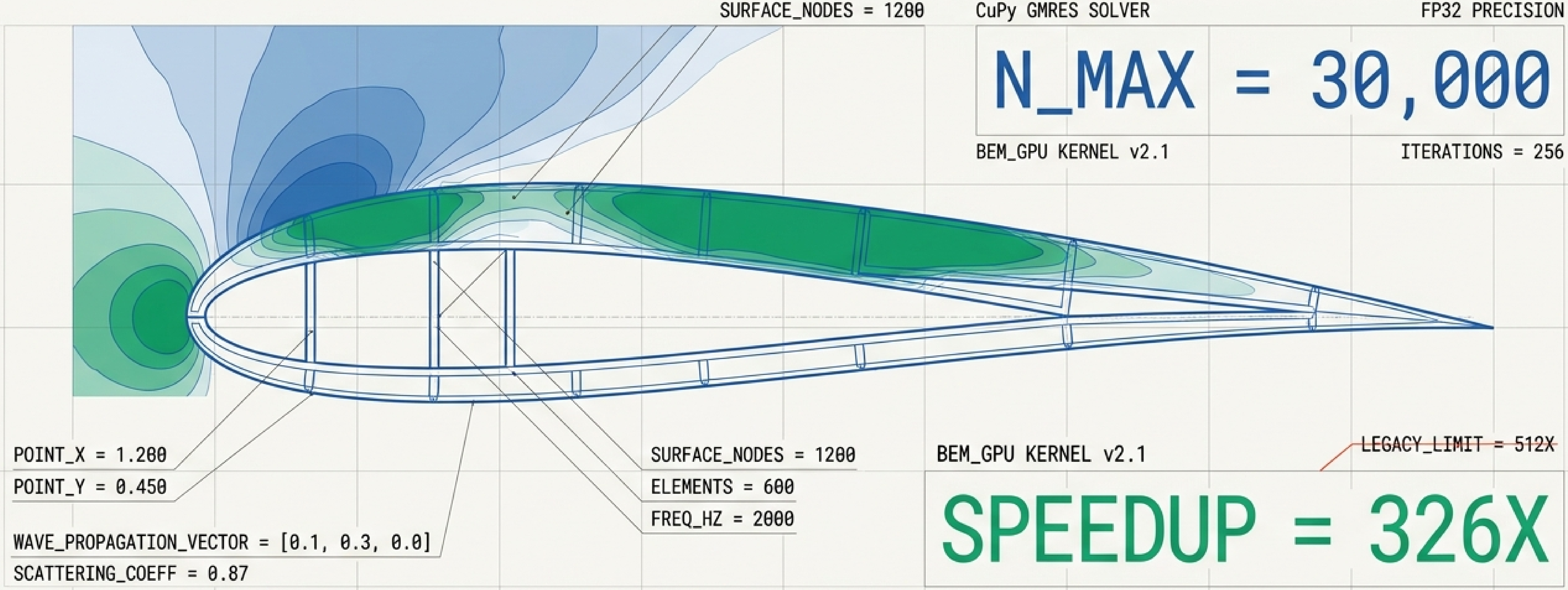
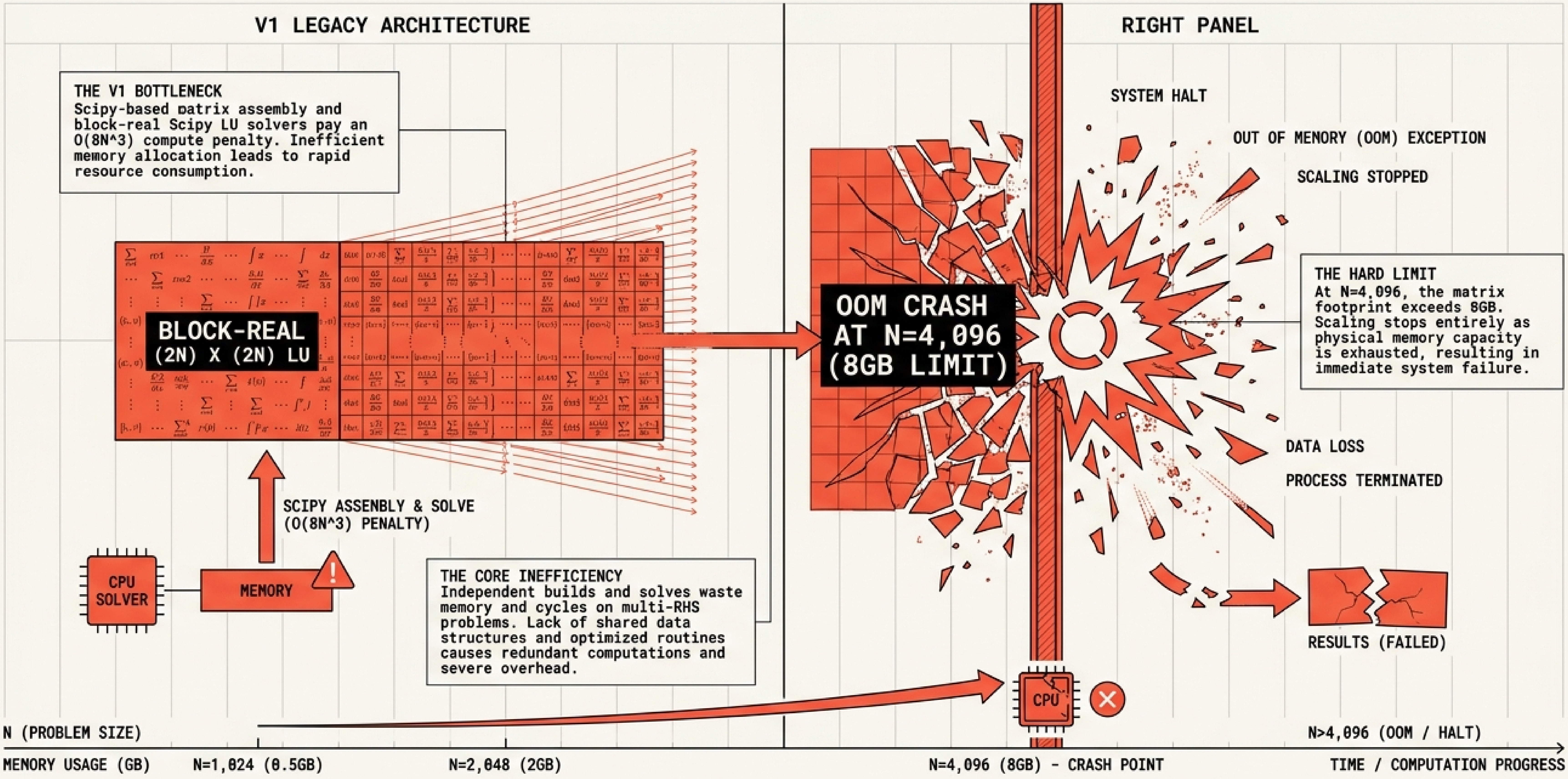
Acoustic Scattering Engine V2

Unlocking massive-scale boundary element method (BEM) workflows on consumer GPU hardware.

Powered by the bem_gpu kernel and CuPy GMRES.



LEGACY CPU SOLVERS COLLAPSE AT THE 8GB MEMORY WALL

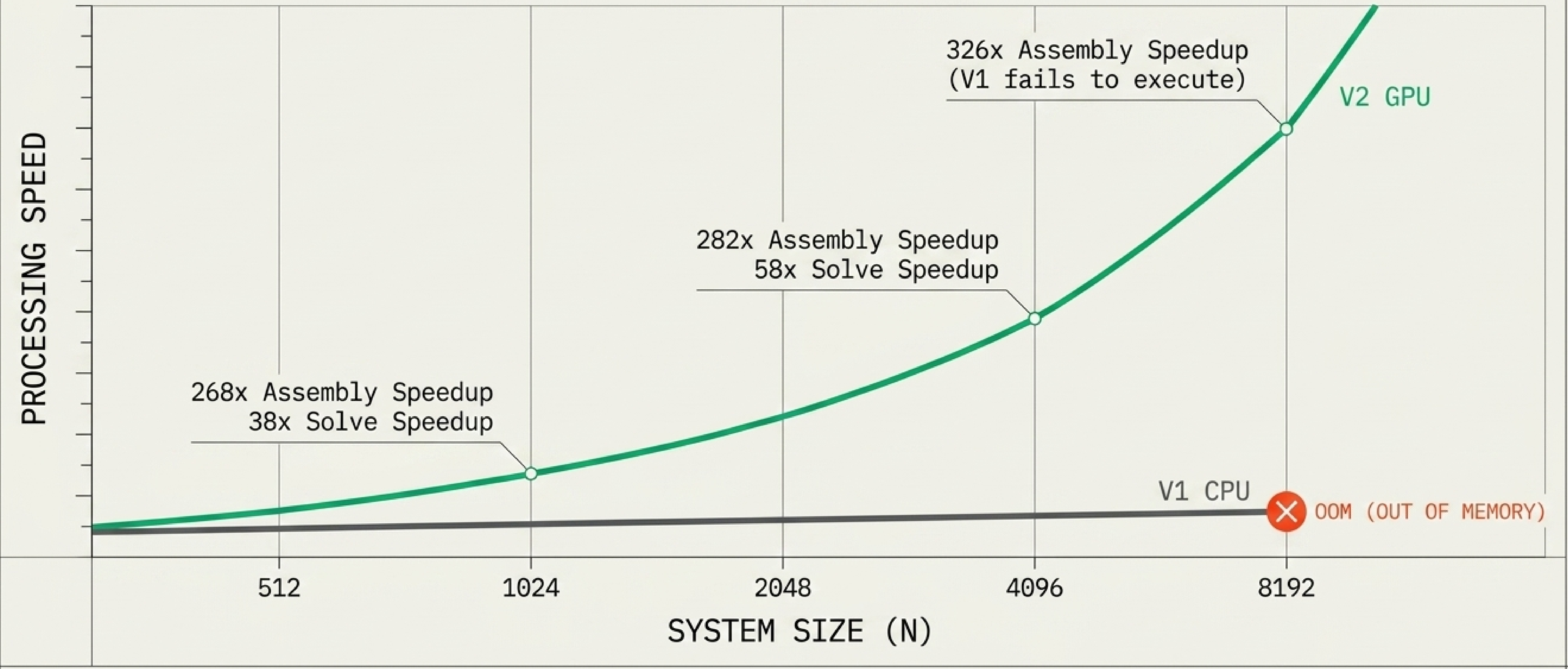


RE-ENGINEERING THE ACOUSTIC BEM STACK FOR GPU SUPREMACY

	LAYER	V1 (ORIGINAL)	V2 (GPU ENGINE)
1	MATRIX ASSEMBLY	CPU scipy.special.hankel1	CUDA Fortran bem_assembly.so
2	SOLVER ENGINE	Block-real (2N)x(2N) Scipy LU	Complex NxN CuPy GMRES
3	SCALING LIMIT	~ 4,096 (OOM)	~ 30,000 (Iterative)

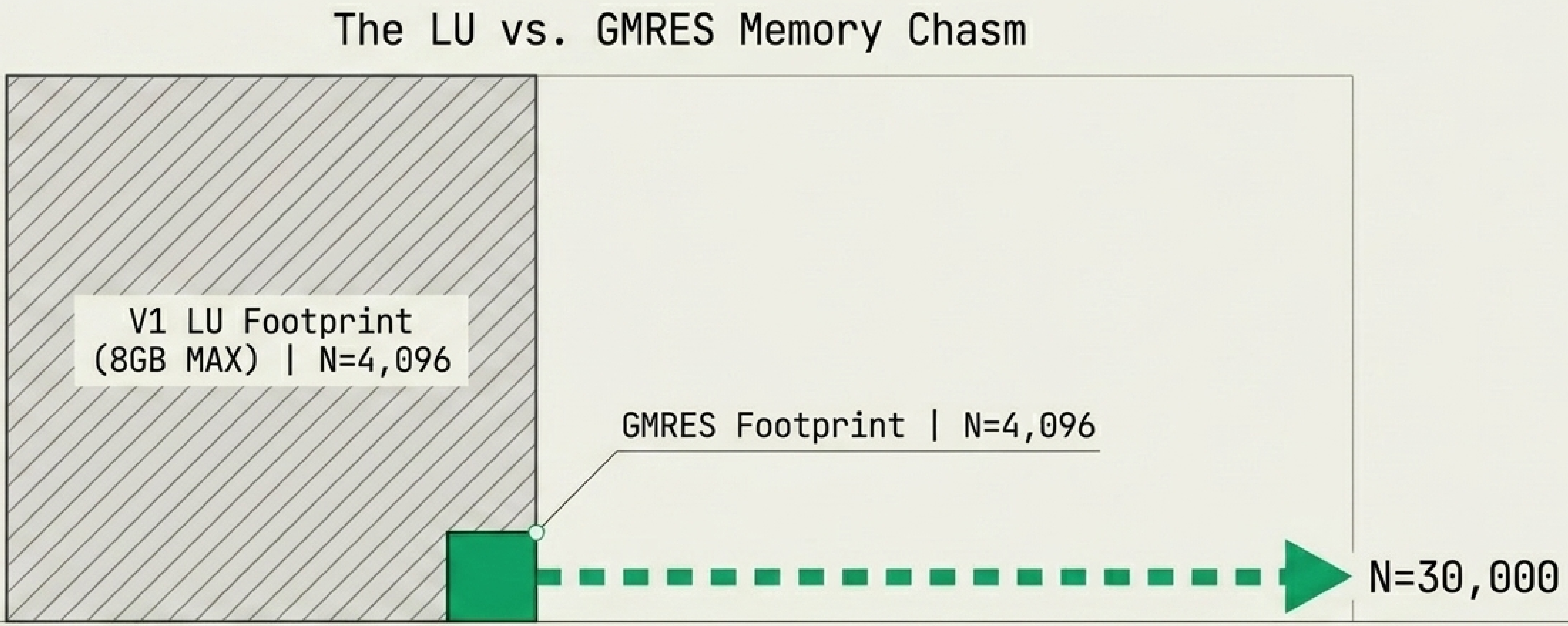
Zero new Fortran written. The 2D acoustic Helmholtz BEM shares the exact Green's function G = (i/4)H_0^(1)(kr)dl with the radar EM BEM. The kernel ports perfectly.

The Kernel Delivers a 326x Speedup and Shatters the N=4,096 Barrier



NotebookLM

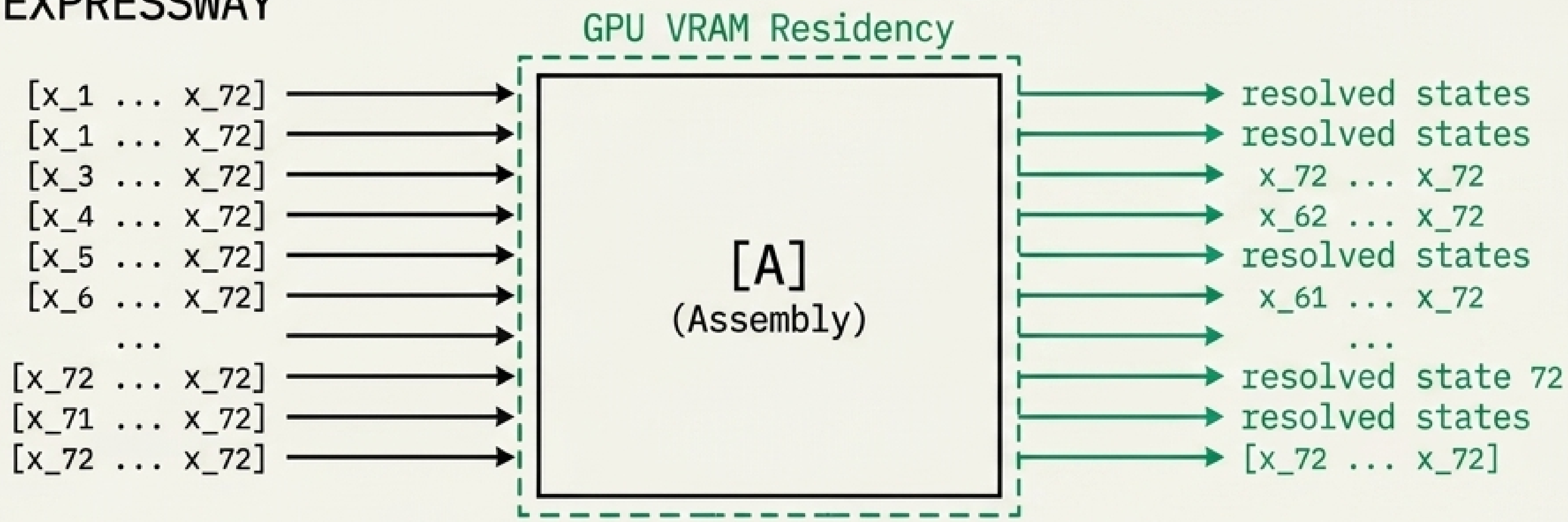
Complex GMRES Iteration Breaks the O(N^2) Memory Wall



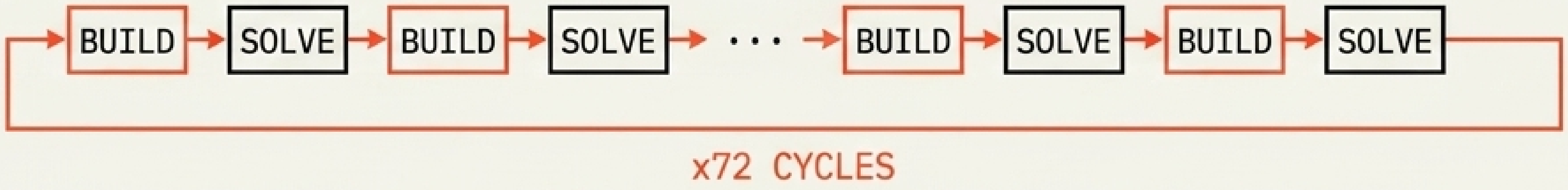
Block-real (2N)x(2N) LU pays 4x the memory price. Complex GMRES on the NxN system smoothly scales to tens of thousands of elements, aided by c64 -> c128 iterative refinement. Condition numbers max at k=683, keeping GMRES highly efficient.

The Multi-RHS Multiplier Maximizes GPU Utilization

V2 EXPRESSWAY



NAIVE APPROACH



Naive processing wastes 71 GPU matrix builds per seed.

V2 loads the matrix once, keeping it resident while executing a unified Multi-RHS solve.

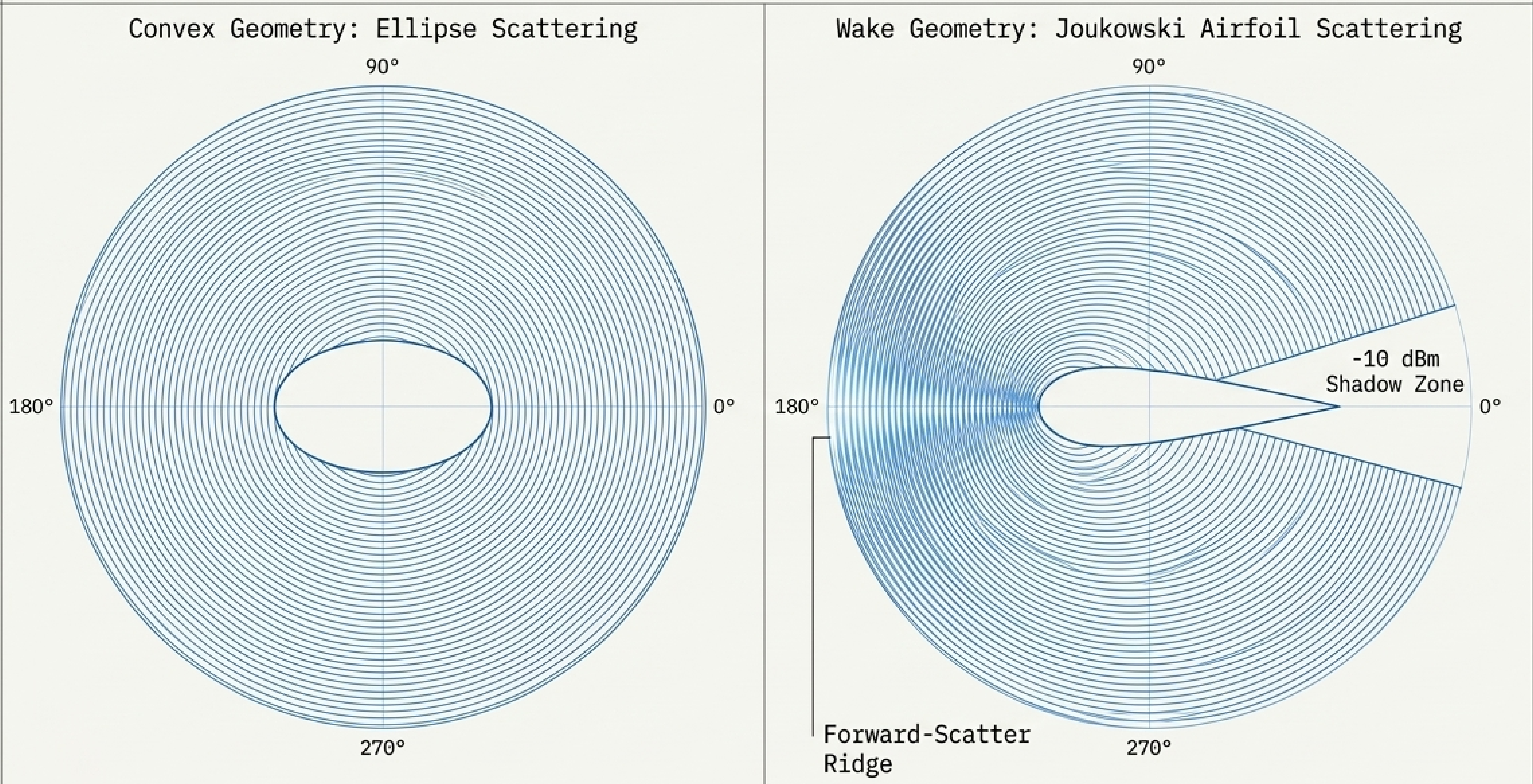
Result: 115,200 solves (N=2,048) execute in just 12 minutes.

The Stealth & Scattering Diagnostic: Convex vs. Wake Geometries

Shape	Monostatic P_{det}	Optimal P_{det}	Bistatic Gain	Shadow Zones
Circle	100%	100%	0 pp	0/72
Ellipse	100%	100%	0 pp	0/72
Joukowski	**100%**	**100%**	**+0.3 pp**	**39/72**
Submarine	100%	100%	0 pp	0/72

Convex shapes radiate energy 360 degrees - bistatic advantage is zero. The Joukowski airfoil operates as an acoustic stealth body, rendering optimally placed receivers blind across 54% of incident directions.

The Bistatic Shadow Cone Escapes Optimal Receivers



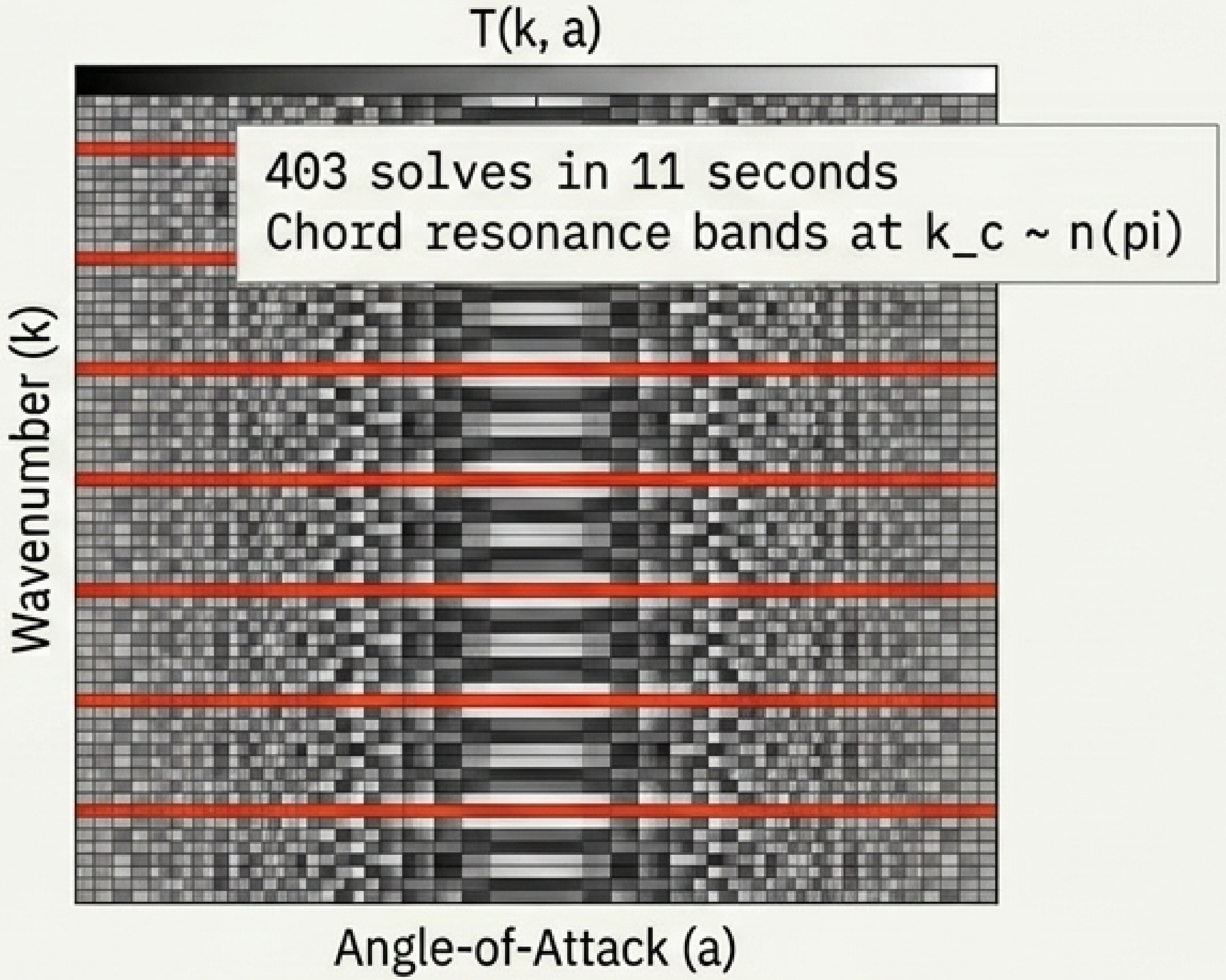
At k=10, the Joukowski shadow depth reaches -10 dBm.

The sharp trailing edge produces narrow, highly coherent nulls.

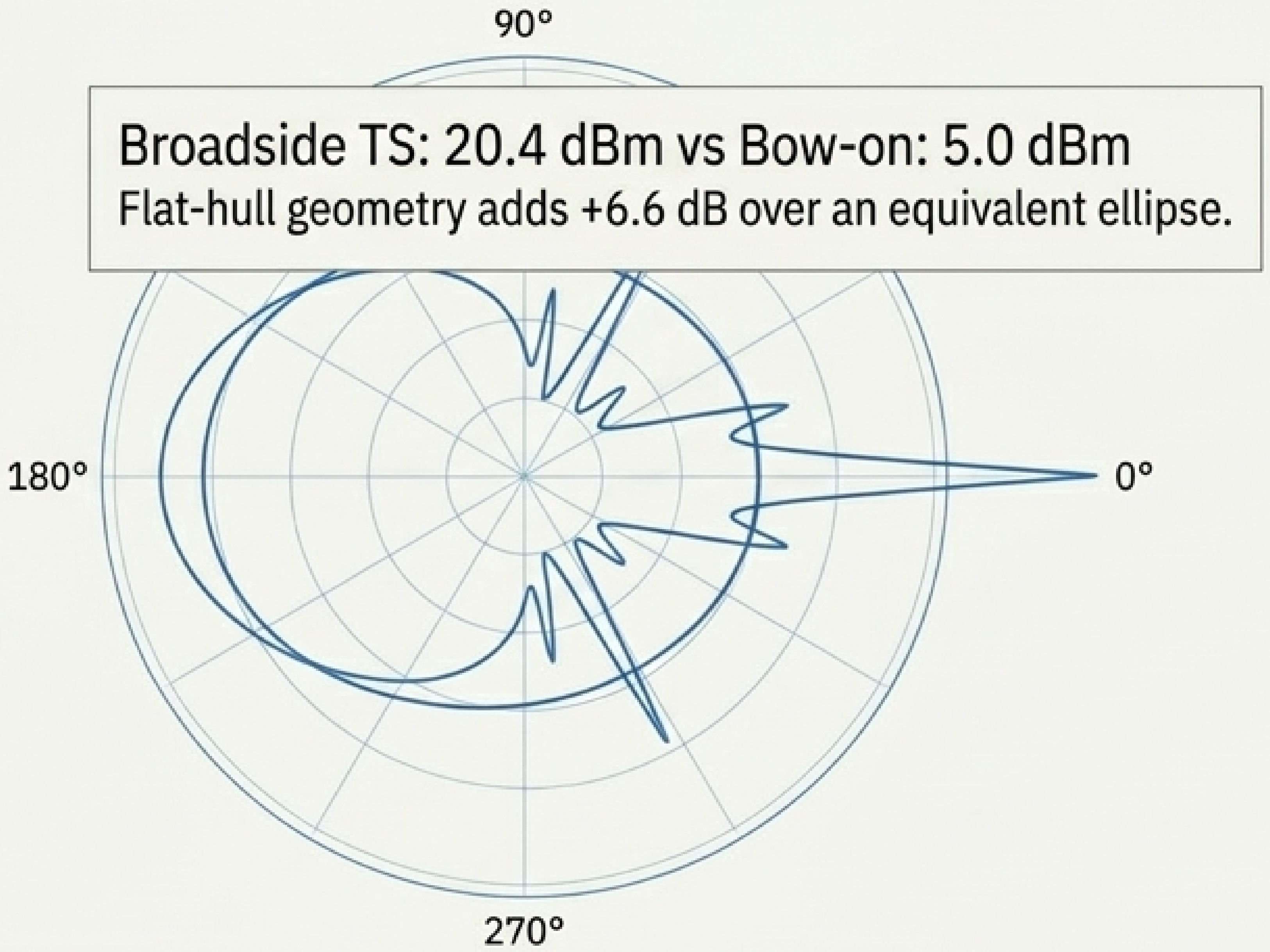
The forward-scatter ridge remains bright, confirming Babinet's theorem across all geometries.

Extreme Compute Density: Generating Fingerprints in Seconds

Joukowski Fingerprint

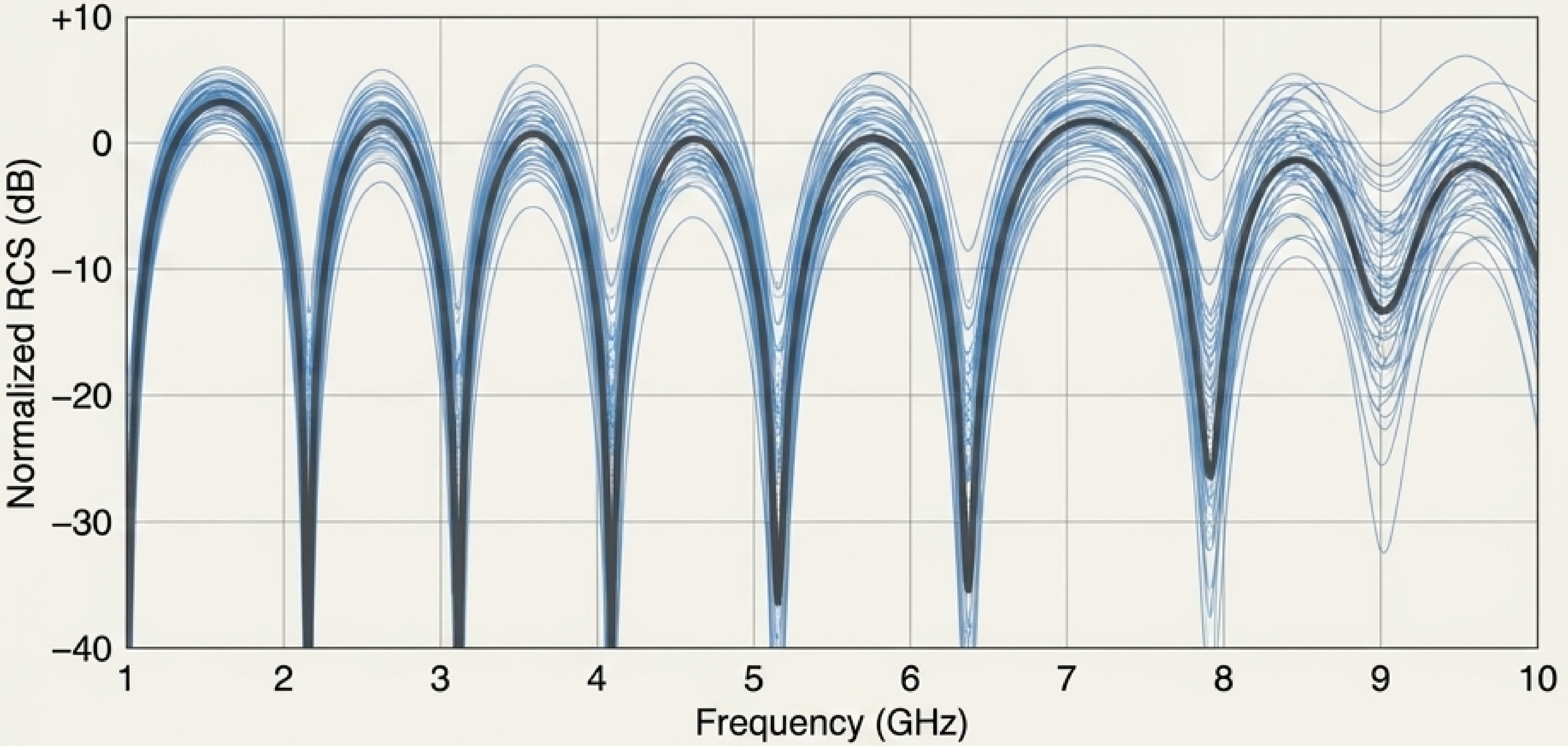


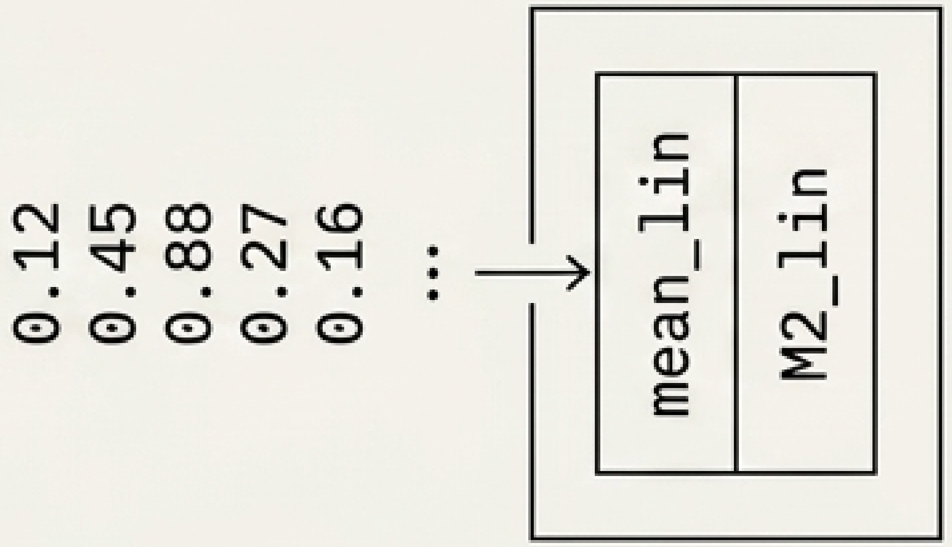
Submarine Sonar Signature



Consumer-grade RTX 4060 hardware now produces parameter-sweep fingerprints in real-time, eliminating overnight compute batches.

Welford Streaming Efficiently Maps Biofouling and Roughness



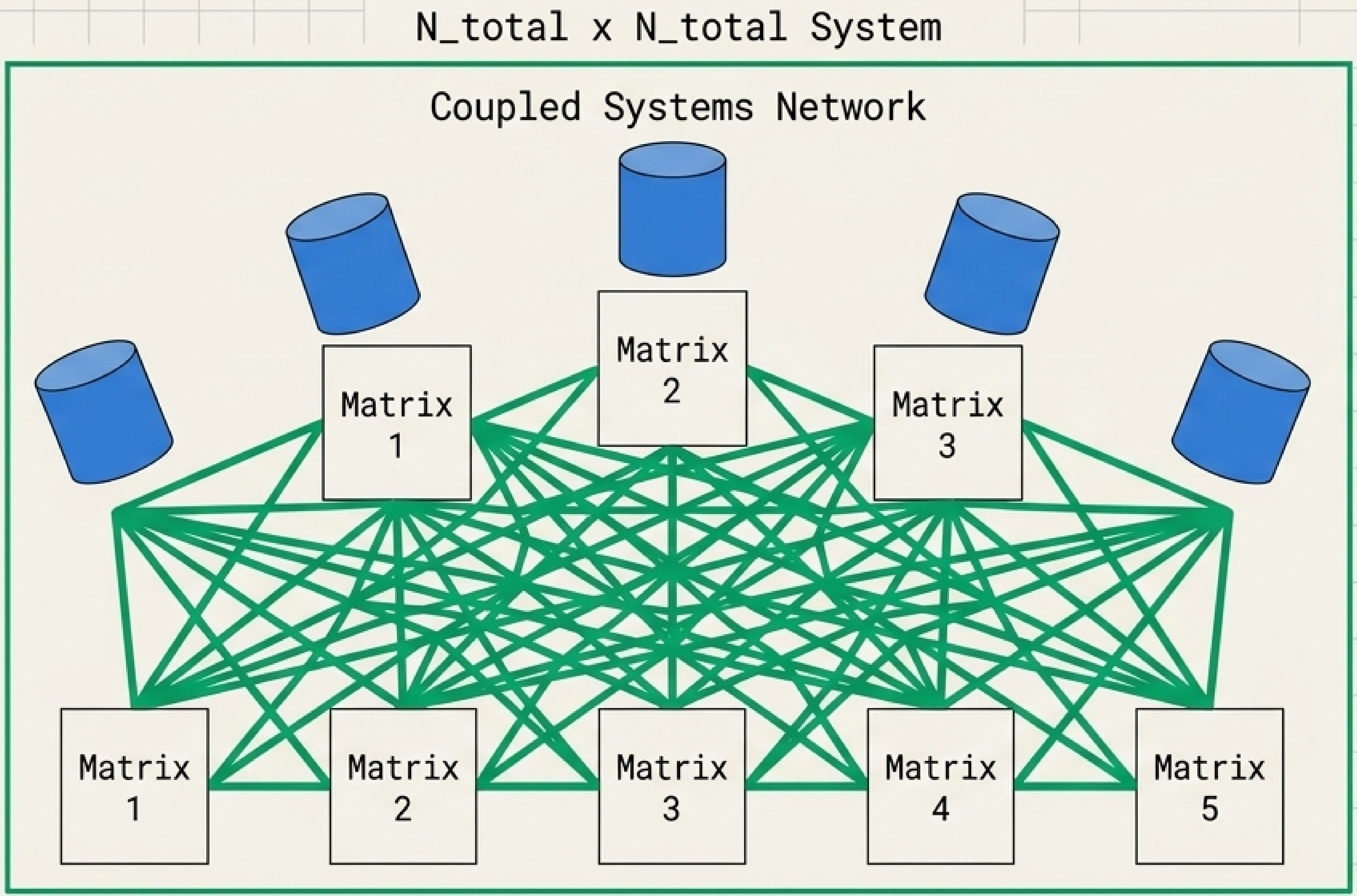


Roughness and biofouling significantly raise the RCS null floor, drastically increasing uncertainty.

10% hull fouling yields stdev=0.40 dB.

Welford streaming calculates rolling statistics in place. Memory requirement remains fixed at two arrays regardless of seed count.

Exact Coupling Networks Correct Single-Scattering Failures



Single-Scattering Failures

Independent single-scattering models fail at sub-wavelength distances (predicting impossible ~200 dB nulls).

FAILURE POINT

Coupled BEM Correction

Coupled BEM properly calculates exact multi-scattering, yielding a median 3.86 dB correction at $d=1\lambda$.

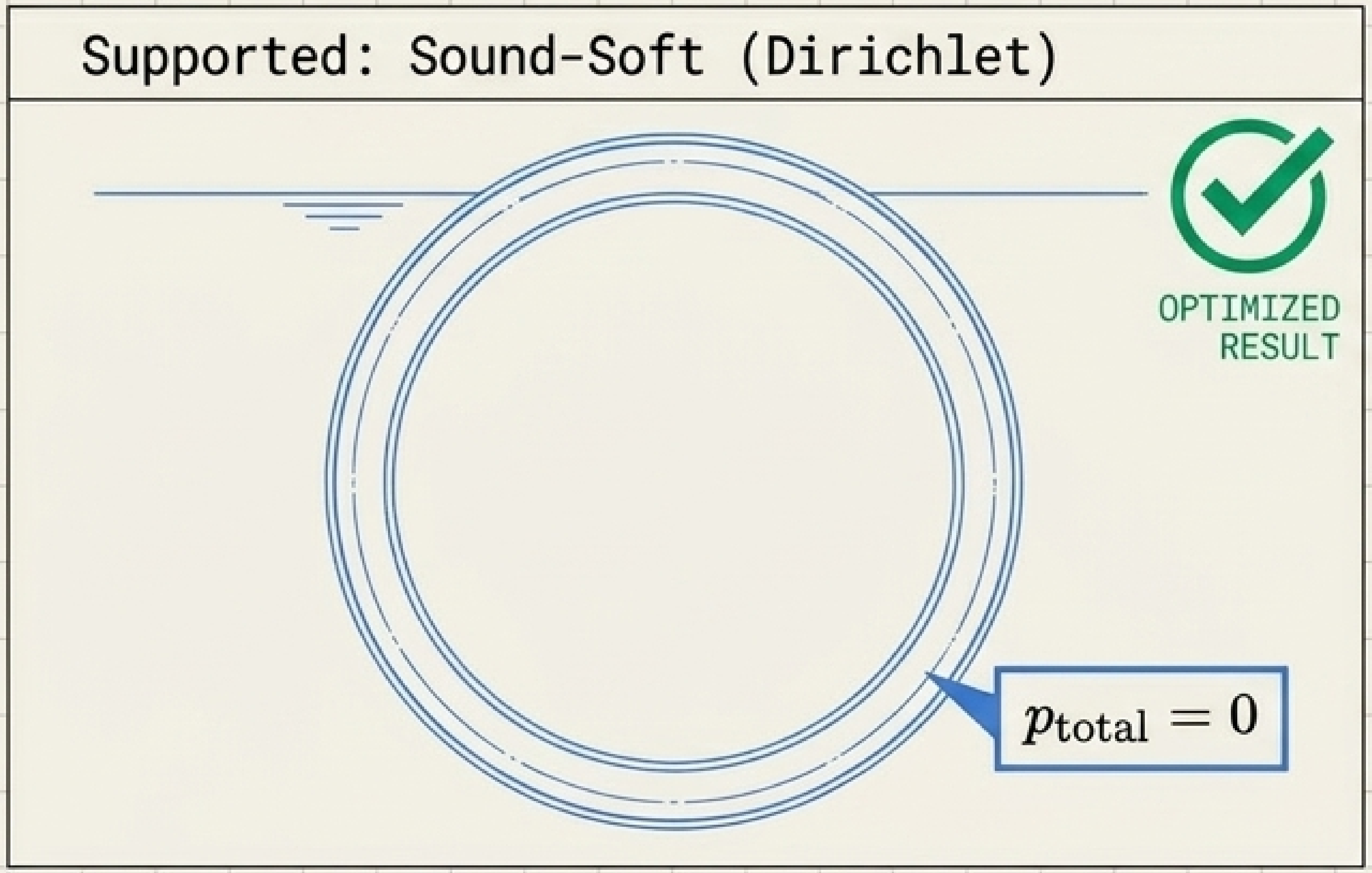
OPTIMIZED RESULT

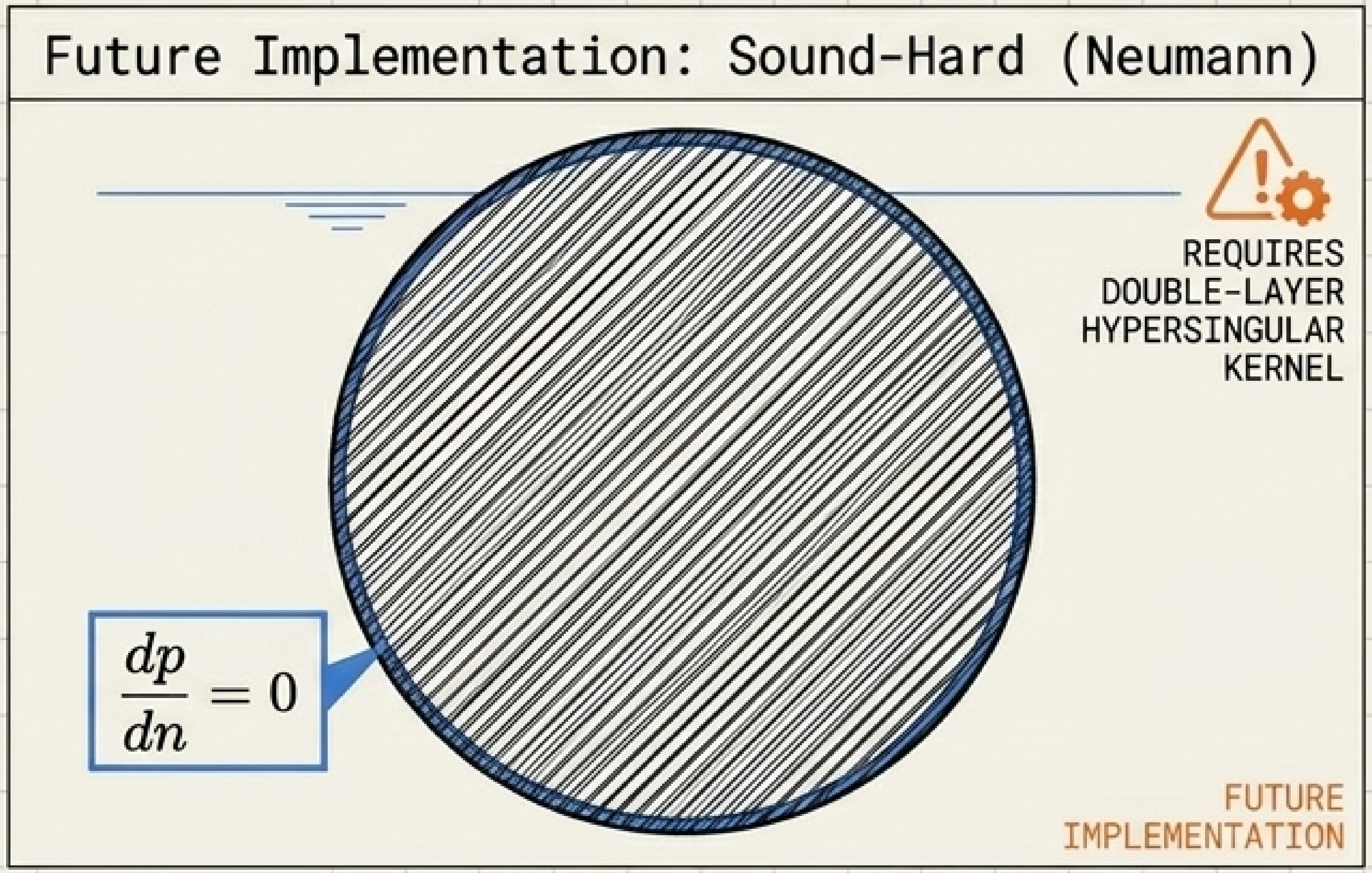
GPU Performance

GPU resolves N_total=960 (5 cylinders) assembly and GMRES in just 20 milliseconds.

REAL-TIME SPEED

Current Physical Boundary Conditions





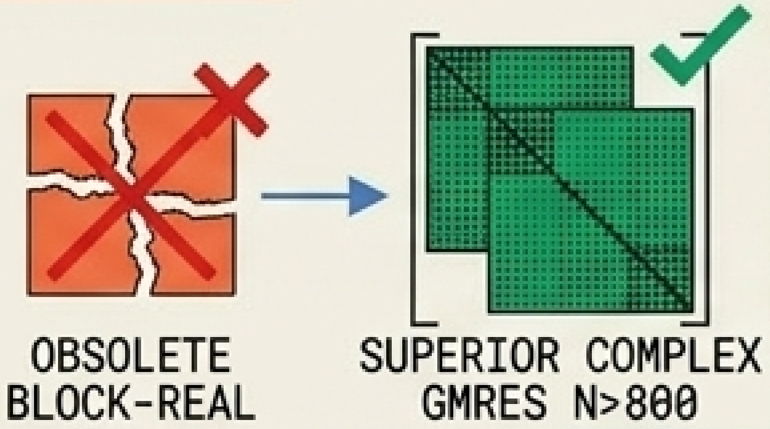
The V2 kernel directly inherits the pressure-release condition. Shape rankings remain internally consistent, but model hollow shells rather than solid metal.

Internal condition-number spikes at high k are brute-forced by GMRES (k_max=683), but extreme scaling will eventually require Burton-Miller regularization.

Universal Master Patterns for 2D Helmholtz Domains

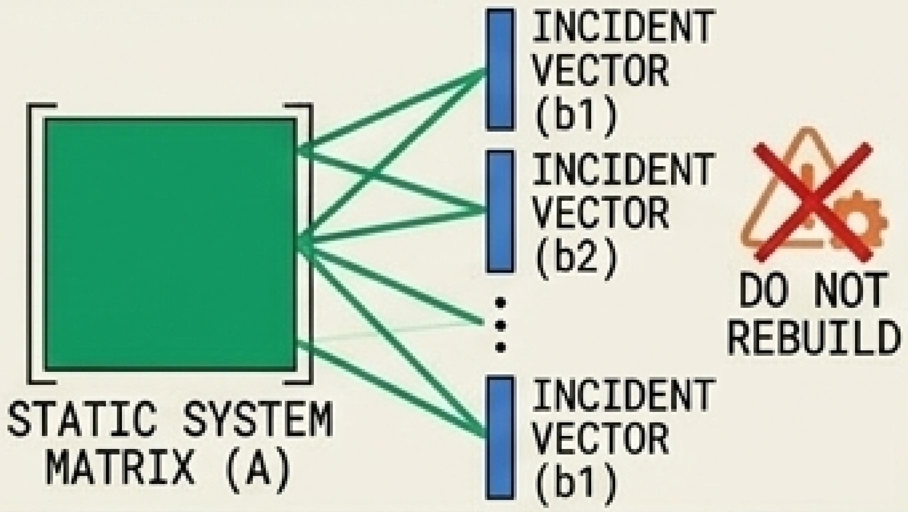
1. Block-Real is Obsolete

Complex NxN GMRES is unconditionally superior above N~800.



2. Multi-RHS is Mandatory

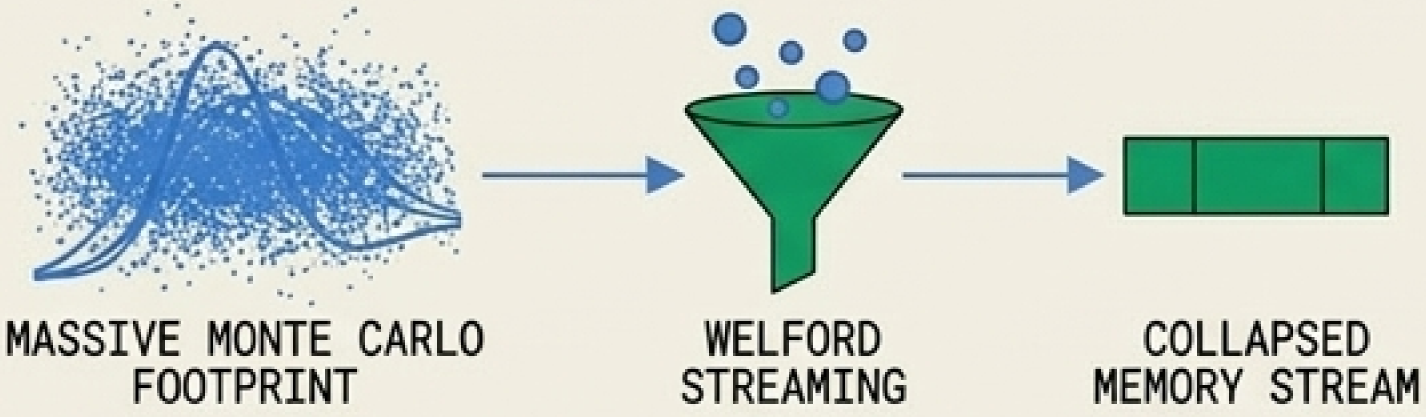
Never rebuild a static system matrix for new incident vectors.



Acoustic V2 Core

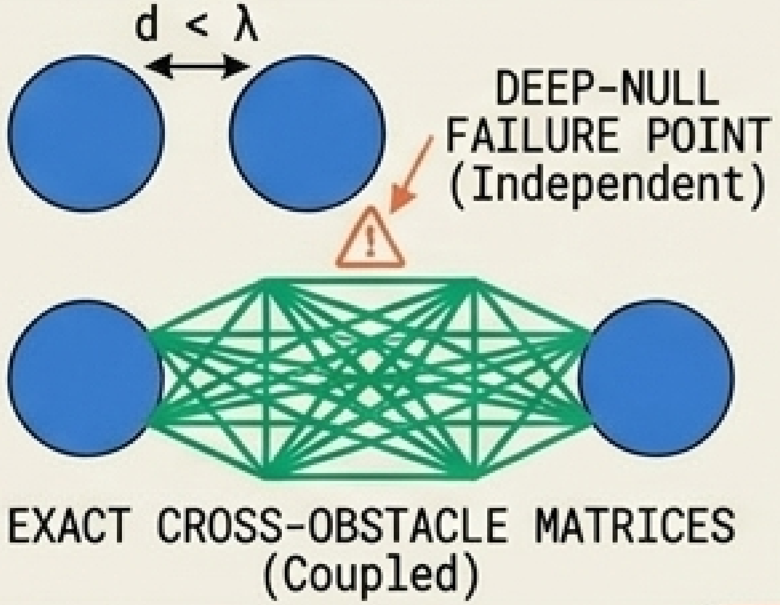
3. Stochastics are Cheap

Welford streaming collapses massive Monte Carlo memory footprints.



4. Coupling is Non-Negotiable

At $d < \lambda$, exact cross-obstacle block matrices are required to prevent deep-null prediction failures.



The Desktop Supercomputer Paradigm

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> EXECUTION TIME : 12m 00s
> TOTAL SOLVES   : 115,200
> HARDWARE       : RTX 4060
> STATUS         : COMPLETE
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Acoustic Scattering V2 proves that high-fidelity computational science is no longer bound by CPU memory walls or HPC queue times.

By treating BEM assembly and iteration as a GPU-native problem, massive-scale statistical, multi-body, and broadband workflows execute in seconds on accessible consumer hardware.